

JASMINE C. OMANDAM

MY ROAD MAP

WEEK 1: BINARY EXPLOITATION & REVERSE ENGINEERING

Binary Exploitation

Challenges:

- Binary Gauntlet 0 ([picoCTF - picoGym Challenges](#))
- Buffer Overflow ([picoCTF - picoGym Challenges](#))
- Heap 1 ([picoCTF - picoGym Challenges](#))
- Heap 0 ([picoCTF - picoGym Challenges](#))
- Format String 0 ([picoCTF - picoGym Challenges](#))

Goals:

- Learn memory layout: stack, heap, registers.
- Understand buffer overflow basics (return address overwrite).
- Explore heap exploitation: chunk manipulation, overwrite validation, brute-force padding strategies.
- Practice debugging with gdb.
- Write exploit payloads using Python + pwntools.
- Document exploit steps clearly (payload length, overwrite verification, flag retrieval).

Reverse Engineering

Challenges:

- Flag Hunters ([picoCTF - picoGym Challenges](#))
- Reverse ([picoCTF - picoGym Challenges](#))
- Transformation ([picoCTF - picoGym Challenges](#))
- Vault Door Training ([picoCTF - picoGym Challenges](#))

Goals:

- Identify strings, functions, and control flow in binaries.
- Practice static vs dynamic analysis with radare2 and IDA Free.
- Extract hidden logic and reconstruct pseudocode.
- Strengthen pattern recognition in obfuscated code (Transformation).
- Decode logic-based puzzles (Vault Door Training) by analyzing condition checks and loops.
- Build confidence in recognizing common reverse engineering tricks (string obfuscation, conditional gates).
- Write concise documentation of findings (pseudocode + explanation of logic flow).

Join the CTF Event: <https://tryhackme.com/love-at-first-breach> 14-17 February

([TryHackMe | Love at First Breach 2026 Training](#))